1.5 Graphs of Sine and Cosine Functions

Name _____ Period ____ Date _____

Pre-Calculus 1.5 · Student sheet

I can…

- find the amplitude and period of a sine or cosine function.
- describe phase shifts and vertical shifts.
- sketch a sine or cosine graph and write its equation.

Vocabulary · fill in the missing word

Amplitude — |a|, half the _____ between the maximum and minimum of y = a sin bx

Period — the _____ of one cycle: 2π/b for y = a sin bx

Phase shift — the _____ shift $\frac{c}{b}$ of y = a sin(bx − c)

Part 1 · Wave sliders

Graph y = a sin(bx − c) + d with sliders.

Slider	What it does
a	
b	
c	
d	

Part 2 · Notes and examples (fill in each note, then work the example)

y = a sin(bx − c) + d: amplitude = |a|, period = _____.

Example 1

Find the amplitude and period.

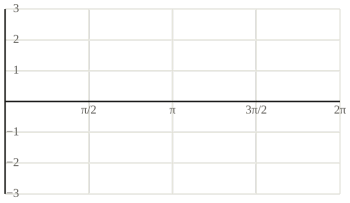
a. $y = 3 \sin 2x$

b. $y = -2 \cos(\frac{x}{2})$

Key points: one cycle splits into _____ equal parts: zero, max, zero, min, zero (for sine).

Example 2

Find the five key points of y = 2 sin x on [0, 2π], then sketch.



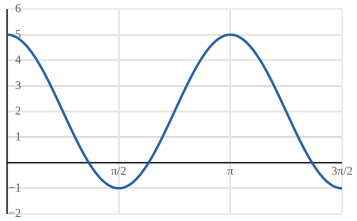
Phase shift: one cycle starts where bx − c = 0 and ends where bx − c = _____.

Example 3

Find the amplitude, period, and the interval for one cycle of $y = (\frac{1}{2}) \sin(x - \frac{\pi}{3})$.

Example 4

Describe $y = 2 + 3 \cos 2x$: amplitude, period, maximum, and minimum.



Example 5

A cosine graph has maximum 4, minimum −2, and period π. Write an equation $y = a \cos bx + d$.

Part 3 · Partner practice

Partner 1

Find the amplitude and period of $y = -4 \sin(\pi x)$.

Partner 2

Find one cycle of $y = \sin(2x - \pi)$.